Data Analytics – Stage One

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# 1. Introduction

This report documents the initial data preparation and exploratory data analysis (EDA) phase for building a movie recommendation system based on the (MovieLens 100K ratings dataset). The primary objective was to move beyond the raw rating matrix by generating sophisticated user and item specific metrics, known as features. Subsequently, Exploratory Data Analysis (EDA) was performed to identify and quantify critical data imperfections, or biases. These biases include user leniency, item popularity, and temporal drift. Identifying these signals helps us build a robust, accurate prediction model by defining the model's architecture and necessary regularization strategies.

# 2. Data Description

The analysis utilized two primary data sources: (ratings.csv and movies.csv). The cleaned dataset contains 100,836 rating records and 9,724 unique movies.

Key columns:

- userId
- movieId
- rating
- timestamp
- title
- genres
- release_year
- genre_count
- rating_year
- user_avg_rating
- movie_rating_count
- movie_age_at_rating
- popularity

The combined dataset covers ratings provided by 610 unique users across 9,724 unique movies. Ratings are on a 0.5 to 5.0 scale.

# 3. Data Cleaning & Preparation

Initial Merging and Type Conversion

The four raw files were loaded, but the 2 major ones (rating and movies) merged on the common key movieId to create a single working DataFrame (df), linking every individual rating event to its corresponding movie attributes.

1. **Timestamp Conversion:** The raw Unix timestamp in ratings.csv was converted into a standard datetime object. This was essential for extracting temporal features and conducting time-series analysis.

2. **Genre Handling:** Movie genres were stored as a pipe-separated string (e.g. “Action|Adventure"). This column was processed to allow for analysis of individual genre popularity and quality.

3. Also checked to confirm there were no duplicated and empty cells in the dataframe.

# 4. Feature Engineering

Six new features were created through extraction, calculation, and aggregation. These features serve as crucial side information to the recommender model, enabling it to learn user and item biases rather than relying solely on the sparse rating matrix. The following features were derived:

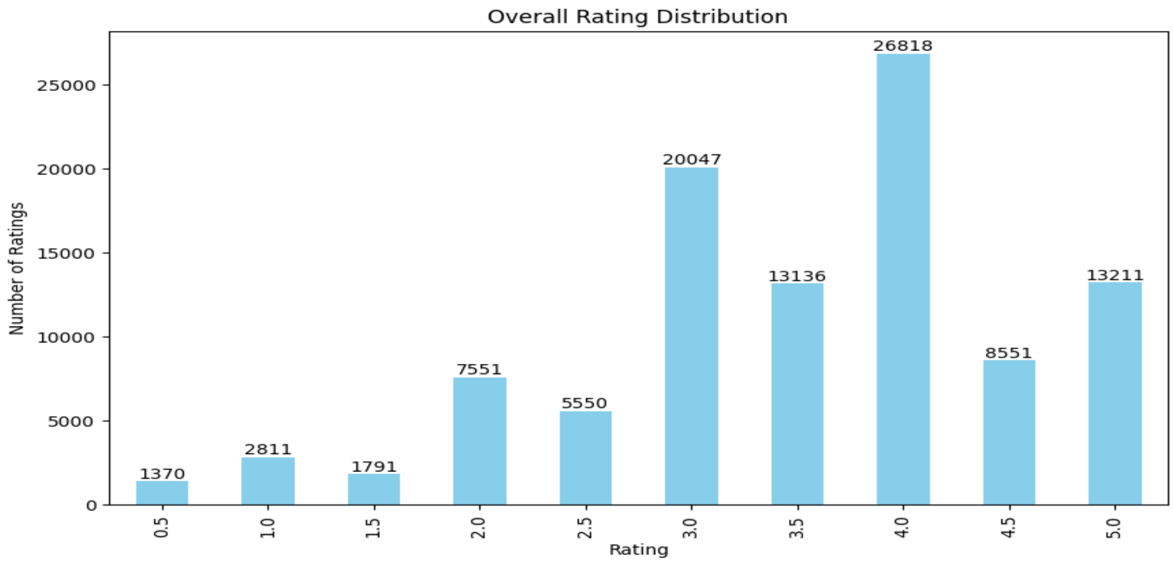
| Feature Name | Creation Method | Usefulness |
|---|---|---|
| release year | Extracted the year from the movie title string. | It is usefulness for analysis on movie age and time trends |
| genre count | Counted the number of genres associated with a movie. | It helps to quantify the movie's niche versus mainstream appeal. Also used to assess diversity in recommendations. |
| rating_year | Extracted the year from the converted timestamp | Helps analyze short-term changes in user behavior and overall rating trends over time. |
| user_avg_rating | Aggregated by userId to find the mean rating. | Measures how lenient or harsh a user’s ratings are, helping adjust for personal rating bias. |
| movie_rating_count | Aggregated by movieId to find the count of all ratings. | Measures both how popular and statistically reliable a movie’s ratings are, helping prevent movies with few ratings from skewing the model. |
| movie_age_at_rating | Calculated as rating_year - release_year. | Shows how old a movie was when rated, helping identify if |

| | | |
|---|---|---|
| | | users prefer new releases or older classics. |

# 5. Exploratory Data Analysis & Visualizations

The EDA focused on six key areas, using the newly engineered features to uncover systematic data biases.

**Question 1:** What is the average rating and do users tend to give high, middle, or low ratings?



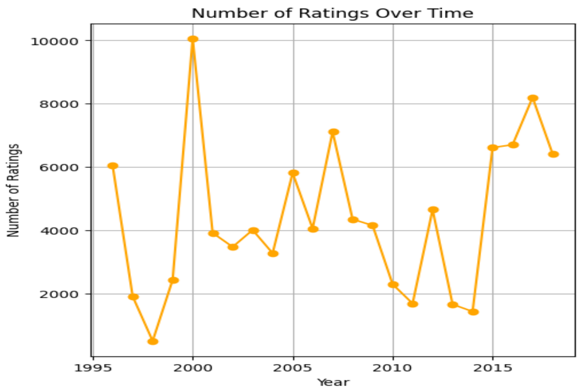

**Insight 1:** Users tend to be generous with ratings, showing a positive bias. The overall mean rating is 3.50 out of 5.0. Furthermore, the mode (most frequent rating) is 4.0, and the majority of ratings fall at 3.0 or higher, indicating that users generally prefer to rate movies they enjoyed or found satisfactory.

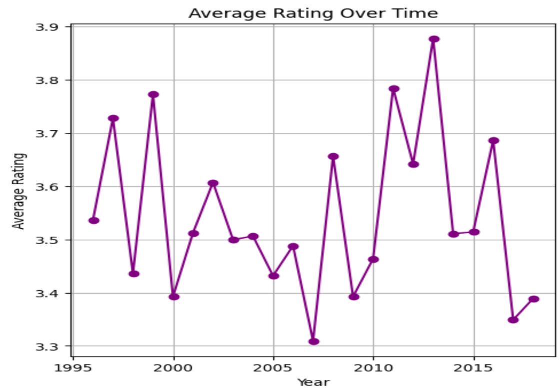
**Question 2:** Do more popular movies receive higher average ratings than less popular movies?

| Popularity Group | Threshold | Average Rating |
|---|---|---|
| Highly Popular | >100 | 3.859 |
| Less Popular | <100 | 3.412 |

**Insight 2:** Popular movies tend to have higher average ratings. Films with 100 or more ratings average about 3.86, compared to 3.41 for less-rated movies. This indicates a popularity bias, meaning widely viewed movies may appear better due to genuine quality or social influence — an important factor to consider when designing a recommendation system.

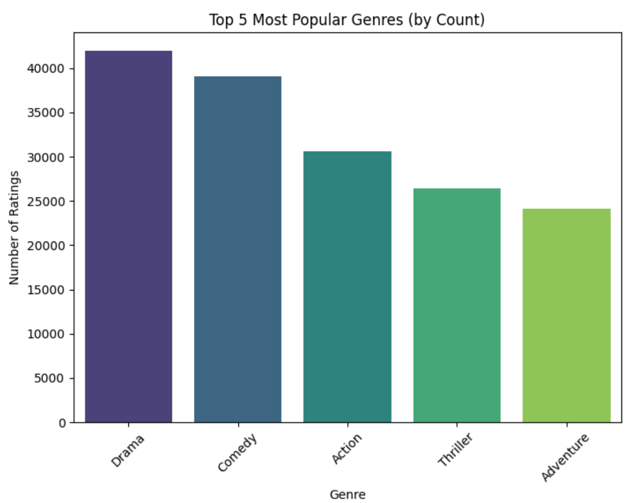
**Question 3:** How have the number of ratings and the average movie rating changed over the years covered in the dataset?

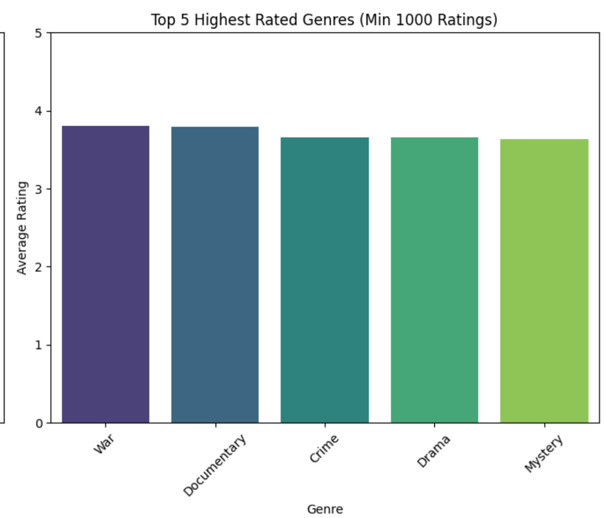



**Insight 3:** User rating activity peaked around 2000 and later stabilized, while average ratings fluctuated over time. The highest average rating occurred in 2013, and the lowest in 2007, showing a year-to-year drift of up to 0.5 points. This time-based variation highlights a temporal bias that should be modeled in any effective recommendation system.

**Question 4:** Which genres gets rated the most, and which receive the highest average ratings?




**Insight 4:** Genre popularity and quality don't always align. While Drama dominates in rating volume and maintains a solid 3.66 average, popular genres like Comedy (3.38) and Action (3.45) rate below the overall average (3.50). In contrast, niche genres such as War (3.81) and Documentary (3.80), though less rated, score highest in user satisfaction. This highlights the need for recommendation systems to balance popularity with quality perception when suggesting content.

**Question 5:** Are newer movies (released in or after 2000) rated differently from older movies (released before 2000)?

| Release Era | Average Rating |
|---|---|
| Old Movies (Pre-2000) | 3.503 |

| | |
|---|---|
| New Movies (Post-2000) | 3.487 |

**Insight 5:** Older movies receive slightly higher ratings than newer ones. Films released before 2010 average 3.50, compared to 3.49 for newer releases, a minimal 0.016-point difference. This suggests no strong "newness bias"; instead, older, well-established movies retain a marginally higher perceived quality.

**Question 6:** Who are the most active users, and how do their average ratings compare to the overall user base?

| User ID | Average Rating |
|---|---|
| 68 | 3.23 |
| 274 | 3.24 |
| 288 | 3.15 |
| 380 | 3.67 |
| 414 | 3.39 |
| 448 | 2.85 |
| 474 | 3.40 |
| 599 | 2.64 |
| 606 | 3.66 |
| 610 | 3.69 |
| **Overall Mean Rating** | **3.50** |

**Insight 6:** Highly active users tend to rate more critically than the average user. Among the ten most active users, average ratings range from 2.64 to 3.69, showing no consistent bias pattern. Notably, several of these users rate below the overall mean (3.50), indicating that the most active contributors are often more critical. Since these users heavily influence recommendation models, it's important to account for their individual rating tendencies.

# 6. Key Findings & How Insights Supports Building Recommendation System

The exploratory insights derived from this analysis provide several foundational directions for developing a robust and data-driven movie recommendation system:

1. **Popularity Bias Adjustment:**
   Popular movies get more ratings and often higher scores, so the system should balance this by also showing less-known but high-quality films to keep recommendations fair and diverse.
2. **Temporal Dynamics and Time Decay:**
   Rating patterns change over time. Adding time-based features like when a rating was made or how old a movie helps the system stay updated with users' changing tastes.
3. **Genre-Specific Weighting:**
   Some genres are popular, while others are rated higher for quality. The model should consider both factors so it can recommend not just popular genres, but also hidden gems that users might love.

4. **User Bias Normalization:**
   Some users are strict, others generous. Adjusting ratings based on each user's personal rating style helps the system understand what they actually like instead of just how they rate.
5. **Age of Movie as a Preference Signal:**
   Knowing whether users prefer new releases or older classics helps personalize recommendations according to each user's viewing habits.
6. **Active User Influence Management:**
   The most active users often rate more harshly. Accounting for this prevents their ratings from unfairly lowering the overall scores of good movies.

# 7. Conclusion & Recommendations

## Conclusion:

The analysis revealed key patterns in user behaviour, movie popularity, and genre preferences. These insights show that factors like time, user bias, and popularity strongly influence ratings and must be considered when building a recommendation system.

## Recommendation:

Future systems should adjust for user bias, include time-based features, and balance popular and niche content to ensure fair, diverse, and personalized recommendations.